

System Architecture — Property Intelligence Infrastructure

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Purpose of the Document

This document describes the system architecture of the Portugal Property Intelligence Infrastructure, including the data platform, analytical engines, and application services used to generate property intelligence insights.

1. Platform Architecture Design

Portugal Property Intelligence Infrastructure

This document defines the **technical architecture of the Property Intelligence Platform**, including the full **data flow, system components, analytical engines, and platform services**.

The architecture follows a **data-centric platform model**, where multiple data sources feed an analytical intelligence layer that powers the buyer and seller platforms.

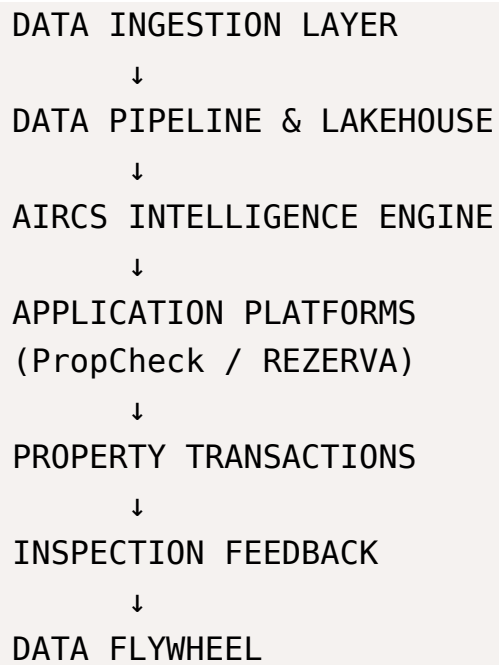
2. System Architecture Overview

The platform operates as a **data intelligence pipeline** where property information flows through several stages: data collection, processing, risk analysis, and application delivery.

Core Data Flow

DATA SOURCES





This architecture enables the platform to **continuously learn from transactions and inspections**, improving its predictive models over time.

This diagram explains the **entire ecosystem at a high level**:

- where the data comes from
- how the intelligence engine works
- how the platforms interact
- how the **data flywheel** improves the system

This diagram provides a high-level overview of the platform architecture, suitable for strategic and technical stakeholders.

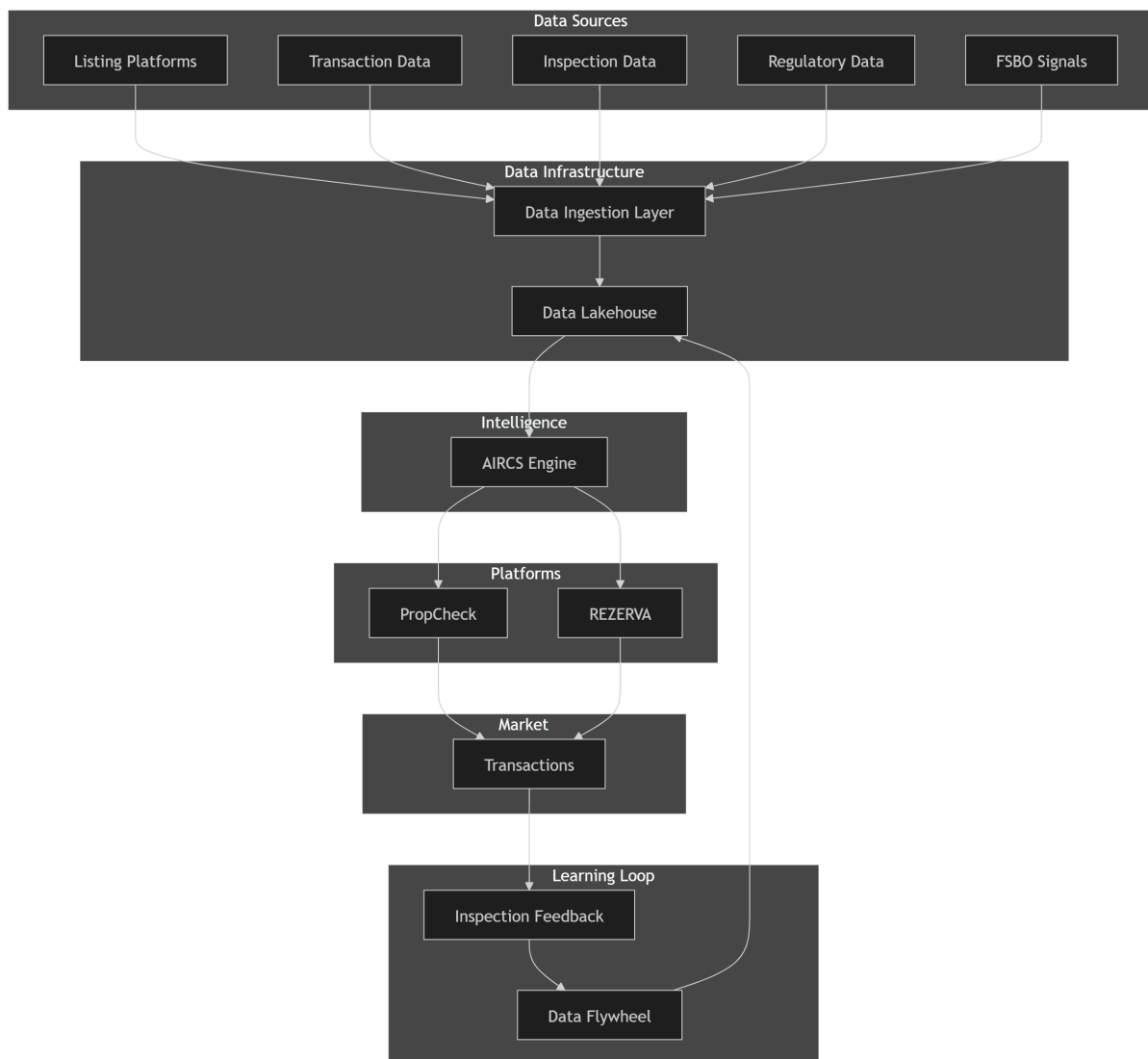


Figure 1 : Platform Ecosystem Architecture

The ecosystem integrates multiple property data sources including listing platforms, transaction datasets, inspection reports, regulatory data, and FSBO signals. These data streams are collected through the data ingestion layer and processed within the platform’s lakehouse architecture.

The AIRCS intelligence engine analyzes the processed data to generate property risk scores and valuation corrections. These analytical outputs power the PropCheck buyer platform and the REZERVA seller platform.

Property transactions generate new inspection and transaction data, which feeds back into the system through the data flywheel. This feedback loop

continuously improves the accuracy of the AIRCS model and strengthens the platform's predictive capabilities.

3. Data Source Layer

The system integrates **multiple heterogeneous data sources** to construct a complete view of property assets.

3.1 Listing Data

These sources provide information about **properties currently offered for sale**.

Examples:

- Idealista
- Imovirtual
- CASAFARI

Collected attributes may include:

- listing price
- property location
- surface area
- construction year
- number of rooms
- energy certificate rating
- listing date

These sources provide **real-time market signals**.

3.2 Transaction Data

Transaction datasets provide **verified historical sale records**.

Primary source:

- SIR transaction database (Confidencial Imobiliário)

Attributes include:

- final transaction price

- transaction date
- property characteristics
- geographic coordinates

This data supports **market alignment analysis and valuation models**.

3.3 Inspection Data

Inspection data is generated internally through **InspectOS**, the inspection infrastructure.

InspectOS collects structured information such as:

- structural defects
- electrical system condition
- plumbing condition
- roof and façade integrity
- energy efficiency indicators
- renovation cost estimates

Inspection records become a **key proprietary dataset** for the AIRCS scoring model.

3.4 Regulatory Data

Regulatory datasets provide insight into **legal compliance and permit status**.

Examples:

- PEPU permit system
- municipal licensing databases
- zoning regulations
- urban planning restrictions

These sources allow the platform to detect **regulatory irregularities**.

3.5 FSBO Signals

The VISBO platform collects signals from **For Sale By Owner (FSBO)** sellers.

These signals include:

- property preparation activity
- documentation uploads
- early seller inquiries

This allows the platform to detect **potential property supply before public listings appear**.

4. Data Ingestion Layer

The ingestion layer is responsible for **collecting data from external and internal sources**.

Data Acquisition Methods

The platform uses several ingestion mechanisms.

API Integrations

Used when providers expose official APIs.

Example:

- CASAFARI
- listing platforms

Web Scraping

Used when APIs are unavailable.

Scrapers extract structured information from listing pages.

Public Dataset Imports

Government datasets are periodically downloaded and processed.

Example:

- municipal permit datasets
- national housing statistics

Internal Data Generation

Inspection data generated by InspectOS is uploaded directly through the inspection app.

5. Data Pipeline and Lakehouse Architecture

After ingestion, the data flows through a **multi-layer data processing pipeline**.

The architecture follows the **Lakehouse model**, which separates raw data from processed analytical datasets.

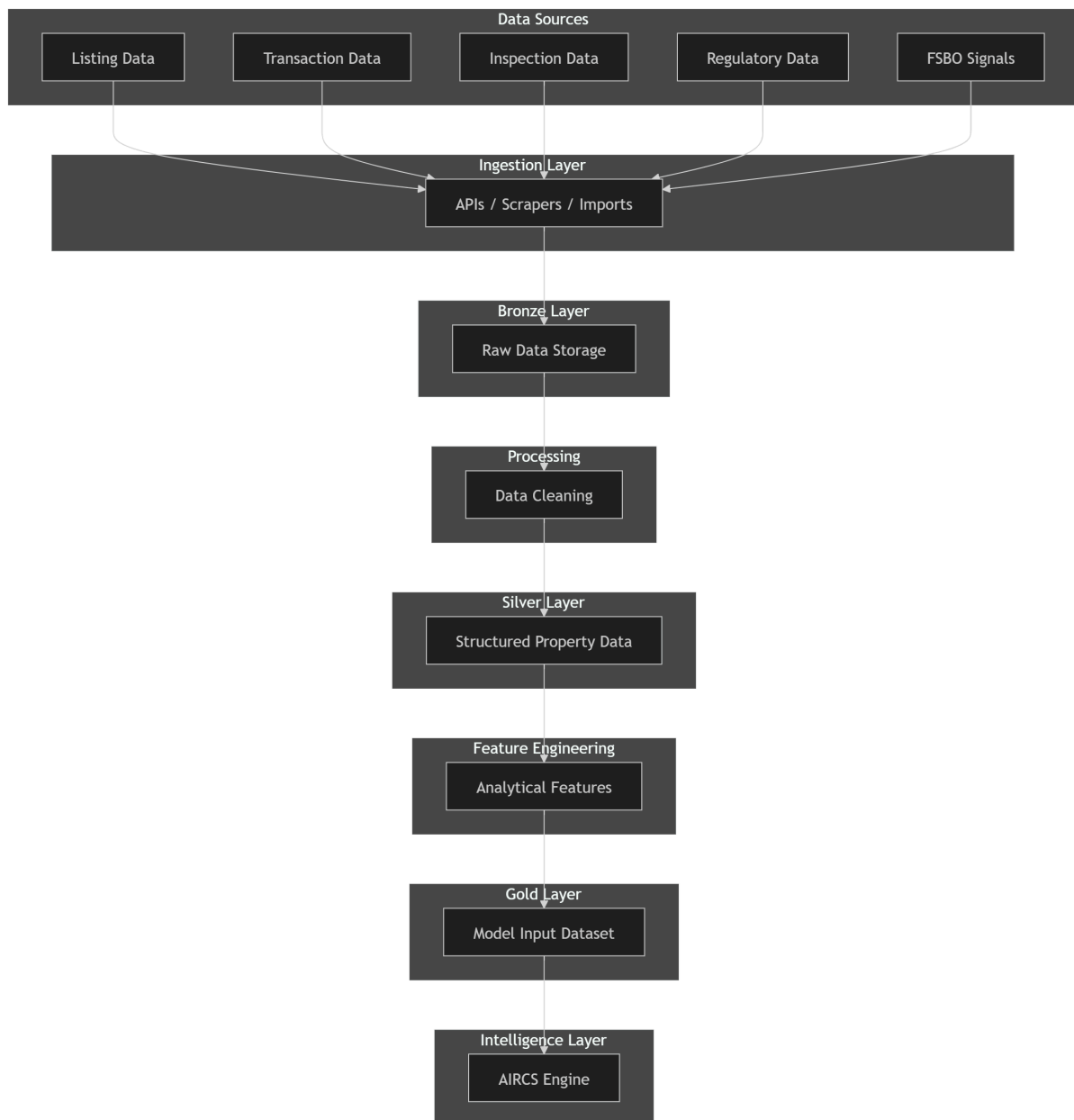


Figure 2 : Data Pipeline Architecture

The data pipeline transforms heterogeneous property datasets into structured analytical datasets used by the AIRCS intelligence engine. Data from listing platforms, transaction records, inspection reports, regulatory databases, and FSBO signals are first collected through the data ingestion layer. Raw datasets are stored in the Bronze layer to preserve original data sources.

Data is then cleaned and normalized in the Silver layer to ensure consistency across different sources. Feature engineering processes convert property attributes into analytical indicators such as structural integrity scores, permit compliance indicators, and price deviation metrics.

The resulting Gold layer contains the final analytical dataset used by the AIRCS engine to compute property risk scores and valuation adjustments.

Bronze Layer (Raw Data)

This layer stores **raw unprocessed data** exactly as collected.

Examples:

- raw listing JSON feeds
- raw inspection reports
- permit data exports

The purpose of this layer is **data preservation and traceability**.

Silver Layer (Cleaned Data)

Data is cleaned and standardized.

Processing tasks include:

- address normalization
- duplicate detection
- missing value handling
- schema standardization

The output is a **consistent property dataset**.

Gold Layer (Analytical Data)

The Gold layer contains **engineered features used by analytical models**.

Examples:

- structural integrity score
- permit compliance score
- price deviation indicator
- energy performance index
- environmental risk index

These features serve as **inputs for AIRCS and valuation models**.

6. AIRCS Intelligence Engine

The AIRCS engine is the **core analytical component of the platform**.

AIRCS converts complex property data into a **single standardized risk score**.

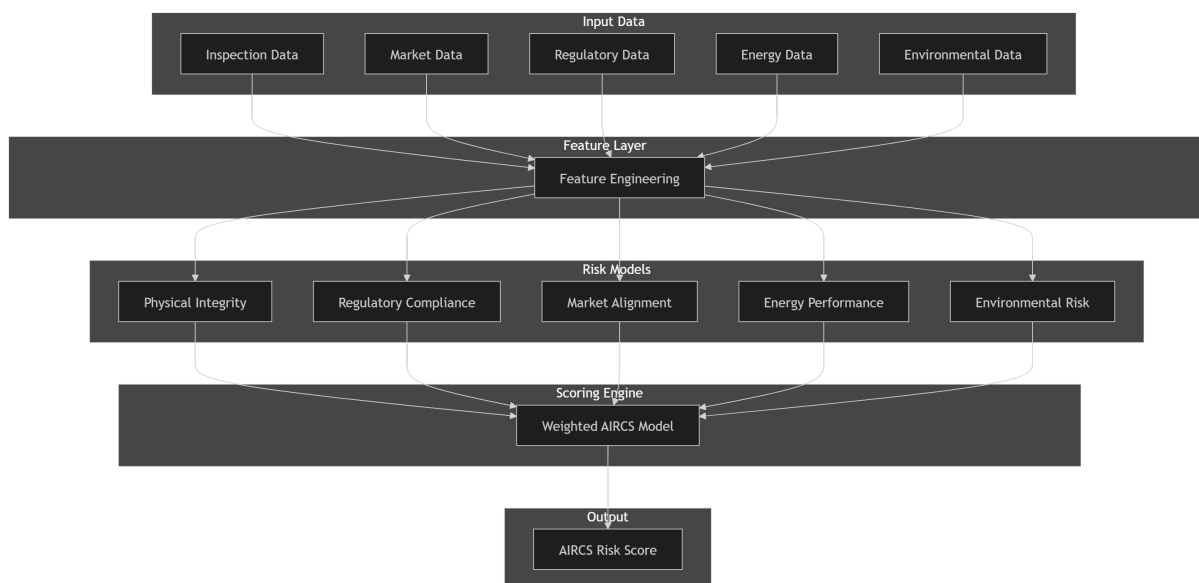


Figure 3 : AIRCS Model Architecture

The AIRCS model integrates multiple sources of property intelligence including inspection data, transaction datasets, regulatory records, energy performance indicators, and environmental risk information. These datasets are processed through a feature engineering layer that transforms raw property attributes into measurable risk indicators.

Each property is evaluated across five risk dimensions: physical integrity, regulatory compliance, market alignment, energy performance, and environmental risk. The outputs of these risk models are combined through a weighted scoring engine that produces a standardized property risk score between 0 and 100.

This unified risk indicator enables buyers, investors, and financial institutions to evaluate property risk in a simplified and interpretable format.

AIRCS Risk Dimensions

The scoring model evaluates five categories of risk.

Dimension	Weight
Physical Integrity	30%
Regulatory Compliance	25%
Market Alignment	20%
Energy Performance	15%
Environmental Risk	10%

AIRCS Scoring Formula

The final score is computed using a weighted model.

```
AIRCS =  
0.30 × Physical Integrity  
+ 0.25 × Regulatory Compliance  
+ 0.20 × Market Alignment  
+ 0.15 × Energy Performance  
+ 0.10 × Environmental Risk
```

Output:

```
AIRCS Score = 0 – 100
```

Example:

```
AIRCS = 72  
Risk Level = Moderate
```

7. Valuation Adjustment Engine (AICF)

AIRCS operates together with the **Asset Integrity Correction Factor (AICF)**.

AICF adjusts property valuation based on inspection findings.

Formula

```
Adjusted Value = Market Value – Estimated Renovation Cost
```

Example:

```
Market Value = €490,000  
Renovation Cost = €55,000  
  
Adjusted Value = €435,000
```

This ensures that valuation reflects **actual building condition** rather than listing price alone.

8. Application Layer

The analytical outputs power two main platforms.

PropCheck — Buyer Intelligence Platform

PropCheck provides decision support for property buyers.

Core features:

- property risk scoring
- valuation analysis
- regulatory compliance verification
- predictive property alerts
- due diligence reports

The platform enables buyers to **evaluate property risk before purchase**.

REZERVA — Seller Capture Platform

REZERVA captures sellers **before properties reach public listing platforms**.

Key functions:

- seller onboarding
- property preparation
- pricing analysis
- buyer matching

This allows the ecosystem to **control supply earlier in the transaction lifecycle**.



Figure 4 : Technical Infrastructure Architecture

The platform architecture is built on a layered system design that separates user interfaces, application services, analytical engines, and data infrastructure. Users interact with the platform through the PropCheck buyer interface, the REZERVA seller platform, and the InspectOS mobile inspection application.

Backend APIs manage user requests and connect the application layer with the intelligence services, including the AIRCS risk scoring engine, the AICF valuation adjustment model, and the predictive sourcing engine.

Operational data is stored in a PostgreSQL database with spatial capabilities provided by PostGIS. Analytical datasets are managed within the platform's lakehouse architecture using Bronze, Silver, and Gold data layers. Data processing pipelines and background jobs are executed through Python-based ETL workflows and message queue systems, while the entire platform is deployed using containerized infrastructure.

9. Transaction Layer

When a property transaction occurs, the system records:

- final transaction price
- inspection results
- negotiation outcomes
- renovation costs

These records provide **valuable ground truth data**.

10. Data Flywheel

Transaction data feeds back into the system.

More Transactions



More Inspection Data



Better AIRCS Calibration



Better Risk Predictions



More Platform Users

This creates a **self-reinforcing data flywheel**.

As the dataset grows, the system becomes **more accurate and difficult for competitors to replicate**.